ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

TECHNOLOGY

PAPER 1 · PRACTICE D

MARKS: 75 **TIME: 1½ hours**

Covers the Grade 7 year: the steps of the design process, a water-tower design, stability and the centre of gravity, frame and shell structures, hardwood and softwood, temporary and permanent joints, tools for cutting, the three classes of levers with moments and mechanical advantage, reverse and parallel motion linkages, circuit symbols, conductors and insulators, electromagnets and energy conversions.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Use a pencil and ruler for drawings. Drawings must be neat and labelled.
4. Show ALL calculations. A non-programmable calculator may be used.
5. Diagrams are NOT drawn to scale.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 The first step of the design process is to …

A make the product. B investigate the problem.
C evaluate the product. D paint the model.

1.2 A structure is more stable when its centre of gravity is …

A high. B on the side. C low. D outside the base.

1.3 Which of these is a shell structure?

A A ladder B A pylon C A bicycle frame D A crash helmet

1.4 Pine is an example of a …

A hardwood. B metal. C softwood. D plastic.

1.5 Which joining method is temporary?

A Nuts and bolts B Welding C Gluing D Riveting

1.6 In a first-class lever, the fulcrum is …

A at one end. B between the effort and the load.
C next to the load only. D not needed.

1.7 A wheelbarrow is an example of a …

A first-class lever. B third-class lever. C linkage. D second-class lever.

1.8 Which material is a good electrical conductor?

A Rubber B Glass C Copper D Dry wood

1.9 An electromagnet is made by winding a coil of wire around a/an …

A iron core. B plastic straw. C copper bar. D glass rod.

1.10 In a torch, chemical energy in the battery changes into electrical energy and then into …

A sound energy. B light energy. C kinetic energy. D nuclear energy.

[10]

QUESTION 2: STRUCTURES

2.1 Write the steps of the design process in the correct order: make, design, evaluate, investigate, communicate (3)

2.2 A village needs a water tower to hold a 1 000-litre tank 4 m above the ground.

2.2.1 Write a design brief for the water tower. (2)

2.2.2 Make a neat, labelled sketch of a frame structure for the water tower. Show how triangles are used to make it rigid. (4)

2.2.3 Suggest a suitable material for the frame and give a reason. (2)

2.2.4 Give TWO ways to make the tower stable so that it does not fall over in strong wind. (2)

2.3 Explain what the centre of gravity of an object is. (2)

2.4 Explain the difference between a frame structure and a shell structure, and give ONE example of each. (3)

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QUESTION 3: MATERIALS AND PROCESSING

3.1 Give ONE difference between hardwood and softwood, and ONE example of each. (3)

3.2 Explain the difference between a temporary joint and a permanent joint, and give ONE example of each. (4)

3.3 Name a suitable tool for EACH job: (a) cutting a straight line in wood; (b) cutting thin sheet metal; (c) smoothing the edge of a wooden shelf. (3)

3.4 Give TWO safety rules to follow when using hand tools. (2)

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QUESTION 4: MECHANICAL SYSTEMS

4.1 Name the class of lever for EACH: (a) a see-saw; (b) a bottle opener; (c) tweezers. (3)

4.2 A crowbar is used to lift a rock. The rock (load) of 600 N is 0,5 m from the fulcrum. The effort is applied 1,5 m from the fulcrum.

4.2.1 Draw a labelled sketch of the crowbar showing the load, effort and fulcrum. (3)

4.2.2 Use the principle of moments (load × load arm = effort × effort arm) to calculate the effort needed. (3)

4.2.3 Calculate the mechanical advantage of the crowbar. Use: MA = load ÷ effort. (2)

4.2.4 How can the effort be made even smaller? (1)

4.3 Explain what a reverse motion linkage does and give ONE example of where it is used. (2)

4.4 Explain what a parallel motion linkage does and give ONE example of where it is used. (2)

4.5 Why is the human forearm (lifting something in the hand) a third-class lever? (2)

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QUESTION 5: ELECTRICAL SYSTEMS

5.1 Draw the circuit symbol for EACH component: (a) a cell; (b) a bulb (lamp); (c) an open switch; (d) a resistor. (4)

5.2 Draw a circuit diagram with two cells, a switch and two bulbs connected in parallel. (4)

5.3 Classify each material as a conductor or an insulator: (a) aluminium; (b) plastic; (c) salt water; (d) cotton. (4)

5.4 Give TWO ways to make an electromagnet stronger. (2)

5.5 Give ONE use of an electromagnet and explain why an electromagnet is better than a permanent magnet for this use. (2)

5.6 Write the energy conversion that takes place in an electric fan. (1)

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TOTAL: 75

END OF QUESTION PAPER